Q1 2021 (01 Sep Shift 2)

$\cos^{-1}(\cos(-5))+\sin^{-1}(\sin(6))-\tan^{-1}(\tan(12))$ is equal to :

(The inverse trigonometric functions take the principal values)

(1) $3\pi-11$

(2) $4\pi-9$

(3) $4\pi-11$

(4) $3\pi+1$

Q2 2021 (31 Aug Shift 2)

The domain of the function

$\mathrm{f(x)}=\sin^{-1}\left(\frac{3\mathrm{x}^2+\mathrm{x}-1}{(\mathrm{x}-1)^2}\right)+\cos^{-1}\left(\frac{\mathrm{x}-1}{\mathrm{x}+1}\right)$ is :

(1) $\left[0,\frac{1}{4}\right]$

(2) $[-2,0]\cup\left[\frac{1}{4},\frac{1}{2}\right]$

(3) $\left[\frac{1}{4},\frac{1}{2}\right]\cup\{0\}$

(4) $\left[0,\frac{1}{2}\right]$

Q3 2021 (27 Aug Shift 2)

Let M and m respectively be the maximum and minimum values of the function $f(x)=\tan^{-1}(\sin x+\cos x)$ in $\left[0,\frac{\pi}{2}\right]$, Then the value of $\tan(\mathrm{M}-\mathrm{m})$ is equal to:

(1) $2+\sqrt{3}$

(2) $2-\sqrt{3}$

(3) $3+2\sqrt{2}$

(4) $3-2\sqrt{2}$

Q4 2021 (27 Aug Shift 1)

If $\left(\sin^{-1}\mathrm{x}\right)^2-\left(\cos^{-1}\mathrm{x}\right)^2=\mathrm{a};0<\mathrm{x}<1,\mathrm{a}\neq0$, then the value of $2\mathrm{x}^2-1$ is :

(1) $\cos\left(\frac{4a}{\pi}\right)$

(2) $\sin\left(\frac{2a}{\pi}\right)$

(3) $\cos\left(\frac{2a}{\pi}\right)$

(4) $\sin\left(\frac{4a}{\pi}\right)$

Q5 2021 (26 Aug Shift 2)

If $\sum_{r=1}^{50} \tan^{-1} \frac{1}{2r^2} = p$, then the value of $\tan p$ is :

(1) $\frac{101}{102}$

(2) $\frac{50}{51}$

(3) 100

(4) $\frac{51}{50}$

Q6 2021 (26 Aug Shift 2)

The domain of the function $\operatorname{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is :

(1) $\left(-1,-\frac{1}{2}\right] \cup (0, \infty)$

(2) $\left[-\frac{1}{2}, 0\right) \cup [1, \infty)$

(3) $\left(-\frac{1}{2}, \infty\right) - \{0\}$

(4) $\left[-\frac{1}{2}, \infty\right) - \{0\}$

Q7 2021 (26 Aug Shift 1)

Let $f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right)$,

$0 < x < 1$. Then :

(1) $(1-x)^2 f'(x) - 2(f(x))^2 = 0$

(2) $(1+x)^2 f'(x) + 2(f(x))^2 = 0$

(3) $(1-x)^2 f'(x) + 2(f(x))^2 = 0$

(4) $(1+x)^2 f'(x) - 2(f(x))^2 = 0$

Answer Key

Q1 (3)	**Q2** (3)	**Q3** (4)	**Q4** (2)
Q5 (2)	**Q6** (4)	**Q7** (3)	

Q1 (3)

$\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$

$\Rightarrow (2\pi - 5) + (6 - 2\pi) - (12 - 4\pi)$

$\Rightarrow 4\pi - 11$

Q2 (3)

$f(x) = \sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right) + \cos^{-1}\left(\frac{x-1}{x+1}\right)$

$-1 \leq \frac{x-1}{x+1} \leq 1 \Rightarrow 0 \leq x < \infty$ ...(1)

$-1 \leq \frac{3x^2+x-1}{(x-1)^2} \leq 1 \Rightarrow x \in \left[\frac{-1}{4}, \frac{1}{2}\right] \cup \{0\}$ ...(2)

(1)&(2)

$\Rightarrow$ Domain $= \left[\frac{1}{4}, \frac{1}{2}\right] \cup \{0\}$

Q3 (4)

Let $g(x) = \sin x + \cos x = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$

$g(x) \in [1, \sqrt{2}]$ for $x \in [0, \pi/2]$

$f(x) = \tan^{-1}(\sin x + \cos x) \in \left[\frac{\pi}{4}, \tan^{-1}\sqrt{2}\right]$

$\tan\left(\tan^{-1}\sqrt{2} - \frac{\pi}{4}\right) = \frac{\sqrt{2}-1}{1+\sqrt{2}} \times \frac{\sqrt{2}-1}{\sqrt{2}-1} = 3 - 2\sqrt{2}$

Q4 (2)

Given $a = \left(\sin^{-1} x\right)^2 - \left(\cos^{-1} x\right)^2$

$= \left(\sin^{-1} x + \cos^{-1} x\right)\left(\sin^{-1} x - \cos^{-1} x\right)$

$= \frac{\pi}{2}\left(\frac{\pi}{2} - 2\cos^{-1} x\right)$

$\Rightarrow 2\cos^{-1} x = \frac{\pi}{2} - \frac{2a}{\pi}$

$\Rightarrow \cos^{-1}\left(2x^2 - 1\right) = \frac{\pi}{2} - \frac{2a}{\pi}$

$\Rightarrow 2x^2 - 1 = \cos\left(\frac{\pi}{2} - \frac{2a}{\pi}\right)$ option (2)

Q5 (2)

$\sum_{r=1}^{50} \tan^{-1}\left(\frac{2}{4r^2}\right) = \sum_{r=1}^{50} \tan^{-1}\left(\frac{(2r+1)-(2r-1)}{1+(2r+1)(2r-1)}\right)$

$\sum_{\mathrm{r}=1}^{50} \tan^{-1}(2\mathrm{r}+1) - \tan^{-1}(2\mathrm{r}-1)$

$\tan^{-1}(101) - \tan^{-1} 1 \Rightarrow \tan^{-1} \frac{50}{51}$

Q6 (4)

$\frac{1+x}{x} \in (-\infty, -1] \cup [1, \infty)$

$\frac{1}{x} \in (-\infty, -2] \cup [0, \infty)$

$x \in \left[-\frac{1}{2}, 0\right) \cup (0, \infty)$

$x \in \left[-\frac{1}{2}, \infty\right) - \{0\}$

Q7 (3)

$f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right)$

$\cot^{-1}\sqrt{\frac{1-x}{x}} = \sin^{-1}\sqrt{x}$

or $f(x) = \cos\left(2\tan^{-1}\sqrt{x}\right)$

$= \cos\tan^{-1}\left(\frac{2\sqrt{x}}{1-x}\right)$

$f(x) = \frac{1-x}{1+x}$

Now $\mathrm{f}'(\mathrm{x}) = \frac{-2}{(1+\mathrm{x})^2}$

or $\mathrm{f}'(\mathrm{x})(1-\mathrm{x})^2 = -2\left(\frac{1-\mathrm{x}}{1+\mathrm{x}}\right)^2$

or $(1-x)^2 f'(x) + 2(f(x))^2 = 0.$